

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of MSc Physics Examination March 2018

PS901 Electromagnetic Theory-II

Date: 26/03/2018 Day: Monday Time: 01:30 PM To 02:00 PM Maximum Marks: 10

MCQ Ans Keys

Q - I Choose the correct answer for the following questions. 10

1. Faraday's law states that
 - (a) A changing electric field produces a magnetic field.
 - (b) A changing magnetic field produces an electric field.
 - (c) A changing magnetic flux produces a magnetic field.
 - (d) Magnetic and electric fields are mutually perpendicular.
2. Which of the following is not Maxwell's equation?
 - (a) $\nabla \cdot E = \rho / \epsilon_0$
 - (b) $\nabla \cdot B = 0$
 - (c) $\nabla \times E = 0$
 - (d) $\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \frac{\partial E}{\partial t}$
3. The electrical equivalence of mechanical mass is
 - (a) Mutual inductance
 - (b) Electric flux
 - (c) Self-inductance
 - (d) Stress tensor divided by current
4. The Poynting's vectors is given by
 - (a) $\frac{1}{\mu_0} (E \times B)$
 - (b) $\frac{1}{\mu_0} (E \cdot B)$
 - (c) $\frac{1}{\epsilon_0} (E \times B)$
 - (d) $\frac{1}{\mu_0} (\nabla \times E)$
5. The boundary conditions for electric field at the interface between two media are
 - (a) $E_1^{\parallel} - E_2^{\parallel} = 0; \epsilon_1 E_1^{\perp} - \epsilon_2 E_2^{\perp} = 0$

(b) $\epsilon_1 E_1^{\parallel} - \epsilon_2 E_2^{\parallel} = 0; \epsilon_1 E_1^{\perp} - \epsilon_2 E_2^{\perp} = 0$

(c) $\epsilon_1 E_1^{\parallel} - \epsilon_2 E_2^{\parallel} = 0; E_1^{\perp} - E_2^{\perp} = 0$

(d) $E_1^{\parallel} - E_2^{\parallel} = 0; E_1^{\perp} - E_2^{\perp} = 0$

6. The diagonal elements of the Maxwell's stress tensor represent

(a) Pressure

(b) Force

(c) Stress

(d) Shear

7. The bound current due to magnetization of a materials is given by

(a) $\nabla \times M$

(b) $\nabla \cdot M$

(c) $\int (\nabla \times M) \cdot da$

(d) $\int (\nabla \times M) \cdot dl$

8. The momentum density stored in the fields of an electromagnetic wave is given by

(a) $\frac{1}{c} S$

(b) $\frac{1}{c^2} S$

(c) $\frac{1}{c^3} S$

(d) $\frac{1}{c^2} S$

9. Refractive index of a medium is defined as

(a) the ratio of velocity of light in space to the velocity of light in the medium

(b) the ratio of velocity of light in the medium to the velocity of light in space

(c) the ratio of density of space to the density of medium

(d) the measure of hardness of the medium

10. An electromagnetic wave travelling from one medium to another gets totally reflected back from the interface when the incident angle is

(a) equal to Brewster's angle

(b) at grazing incidence

(c) equal to 60°

(d) equal to 45°

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Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Answer the following questions as directed **10**

1. Write down the Maxwell's equations in space. **2**
2. A square loop of wire, with sides of length a , lies in the first quadrant of the xy plane, with one corner at the origin. In this region there is a nonuniform time-dependent magnetic field $B(y, t) = ky^3t^2\hat{z}$ (where k is a constant) . Find the emf induced in the loop. **2**
3. Write down the three famous experiments performed by Faraday that connected the magnetic field and electric field together. **3**
4. Write down the continuity equations of electric and magnetic fields in case of linear media and at a surface that carries charge density σ or current density K . **3**

SECTION – II

Q-III Answer the following questions as directed **15**

1. Derive Fresnel's equations for an electromagnetic wave travelling from one medium to another medium separated by an interface. Assume that the wave is incident at angle to the normal to the interface and the polarization of the wave is in the plane of incidence. **5**

OR

A long coaxial cable of length l , consists of an inner conductor (radius a) and an outer conductor (radius b). It is connected to a battery at one end and a resistor at the other. The inner conductor carries a uniform charge per unit length λ , and a steady current I to the right; the outer conductor has the opposite charge and current. What is the electromagnetic momentum stored in the fields?

2. Suppose the electric and magnetic fields of a plane monochromatic electromagnetic wave in vacuum are given by 5

$$E(\vec{r}, t) = \vec{E}_0 \exp[i(\vec{k} \cdot \vec{r} - \omega t)]$$

$$B(\vec{r}, t) = \vec{B}_0 \exp[i(\vec{k} \cdot \vec{r} - \omega t)]$$

Where, E_0 and B_0 are constant vectors that define the maximum size and directions of electric and magnetic vectors respectively.

- (a) plug this pair of plane wave solutions into the Maxwell's equations in vacuum. Write the four Maxwell's equations as algebraic relationships between the k –vector and the constant vectors E_0 and B_0 .
(b) What is the meaning of the direction of the k –vector? What is the meaning of the magnitude of the k –vector.

3. The Maxwell stress tensor is given by the following expression: 5

$$T_{ij} = \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right)$$

Write down all the components of the tensor in terms of the components of the electric and the magnetic fields. Obtain the force acting on the northern hemispherical bowl of a uniformly charged solid sphere of charge Q and radius R .

OR

Show that the electromagnetic waves are transverse in nature. Obtain a relation between the electric field, magnetic field and the wave vector.